

Dev Sanghvi

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Education

Rice University

Master of Computer Science (AI Specialization)

– GPA: 3.89/4.00

Aug 2025 - Aug 2026

Houston, TX

Adani University

B.Eng. in Computer Science and Engineering (AI-ML)

– CPI: 9.34/10.00; SPI: 10.00/10.00 (8th Sem)

Sep 2021 - Apr 2025

Ahmedabad, Gujarat, India

Experience

Team Leader

Adiance Technologies Pvt. Ltd.

- Led sub-second **real-time activity tracking** on edge video by architecting and deploying **Florence2** and **TimeSformer** pipelines.
- Cut authentication latency by 40% via **C++ ONNXRuntime** face verification, enabling secure on-device access.
- Collaborated with stakeholders in **20+ countries** to tailor Edge AI CCTV deployments to diverse hardware and customer requirements.

Jan 2025 - Jul 2025

Ahmedabad, India

Machine Learning Researcher

Indian Space Research Organisation (ISRO)

- Engineered an **OCR + transformer** security pipeline for offline, privacy-preserving multilingual malicious email detection, reaching **98%** accuracy.
- Improved detection reliability by reducing false positives **70%** compared to rule-based baselines.

Jan 2025 - Apr 2025

Ahmedabad, India

AI Solutions Developer

VMukti Solutions

- Deployed **intelligent video analytics** pipelines across large-scale surveillance systems to enable real-time anomaly detection and activity tracking.
- Engineered and optimized edge inference for **10K+ IoT cameras**, cutting model size **60 to 80%** and speeding inference **2 to 5x** while preserving accuracy.

May 2022 - Jan 2025

Ahmedabad, India

Projects

DP-accurate DAU/MAU Counter Under Deletions | Personal Project

- Designed a deletion-aware distinct-counting pipeline with KMV sketches, tombstone propagation, and RDP accounting for GDPR-compliant DAU/MAU analytics.

Sep 2025 - Dec 2025

Editability and Faithfulness Metric for FEVER | Personal Project

- Benchmarked EF on FEVER with gpt2 and ROME edits across diverse valid examples, comparing target and control flips to quantify editability-faithfulness.

Sep 2025 - Dec 2025

Activity Identification and Triggering System | Personal Project

- Built a video activity detection pipeline with **YOLOv11**, **TimeSformer**, FastAPI, and a React dashboard, reaching about **95%** validation accuracy with WebSocket updates and configurable thresholds.

Jan 2025 - Apr 2025

Technical Skills

- **Languages:** Python, C++, C, JavaScript, TypeScript, SQL, Go
- **ML/AI:** PyTorch, TensorFlow, Keras, OpenCV, Tesseract, EasyOCR, ONNX, TensorRT, QAT-PTQ, Transformers
- **Computer Vision/Video:** YOLO, TimeSformer, DeepSort, RTSP/RTMP, video analytics
- **Web/Systems:** FastAPI, Flask, React, WebSockets, Pandas, NumPy, REST APIs, PostgreSQL, Docker, Git, Linux

Publications

Vision Transformer for Accelerated Design of Very Low-frequency Metasurface Absorber

- Applied ViT architecture to optimize metasurface absorber designs, trimming training time by **40%**.
- Presented at: **CASML 2024, IISc Bangalore**

2024

Awards

- | | |
|---|----------|
| 1st Place, E-Yantra Robotics Workshop | Dec 2022 |
| – Directed the team to design and program robotic arms, winning top honors at IIT Bombay. | |
| Winner, AU-IQM Hackathon | Nov 2022 |
| – Secured 1st place among 50+ teams for an AI-powered data classification solution. | |

Leadership & Involvement

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| Vice President | Mar 2023 - Aug 2024 |
| <i>Adani Student Programming and Development Club (ASPDC)</i> | |
| – Orchestrated 10+ workshops/hackathons and mentored 250+ students, building campus ML/AI culture. | |
| Treasurer | Mar 2024 - Mar 2025 |
| <i>ACM, Association for Computing Machinery</i> | |
| – Directed finances and event planning, supporting student computing initiatives. | |
| Coordinator | Jun 2023 |
| <i>Global Education Forum, Adani University</i> | |
| – Coordinated logistics and volunteers for a 300+ attendee event, ensuring smooth operations. | |

Certifications

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| Microsoft Azure Fundamentals: Cloud Concepts | Mar 2023 |
| Python 101 for Data Science, Cognitive Class | Mar 2023 |
| Jira (Coursera) | Feb 2024 |